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Time series analysis and prediction of PM2.5 pollution concentration in Igdir province with deep learning models

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HIGHLIGHTS

- Comprehensive dataset integrates air quality, meteorological, satellite and time data.
- Missing data imputation and outlier correction applied for data quality.
- Performance of different deep learning architectures (e.g., LSTM, GRU) assessed.
- Models effectively capture key temporal patterns and fluctuations in PM2.5.

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ABSTRACT

Air pollution is a global problem that causes serious environmental and health problems, especially in regions where industrialization and urbanization are intense. Igdir province, located in the eastern part of Türkiye, is one of the regions where air pollution is intensely observed due to its geographical structure and meteorological characteristics. This study evaluated deep learning models for predicting PM2.5 levels using data from national monitoring networks, meteorological services, and NASA POWER. Preprocessing included interpolation, outlier correction, and min–max normalization. LSTM, GRU, Bi-LSTM, Bi-GRU, CNN-LSTM, and CNN-GRU models were tested across 8, 24, and 72 h windows. The GRU model achieved the best performance in short-term (8 h) predictions with MAE=9.93 and R^2=0.944 values. The LSTM model reached the best predictive performance for the 24 h window with MAE=9.65 and R^2=0.949, while for the 72 h window, the BiLSTM model outperformed the others. In terms of predicting peak values, the CNN-LSTM model stood out, achieving RMSE=28.16, R^2=0.792, and MAE=22.45 in the 8 h window. These findings highlight deep learning's efficacy for air pollution forecasting and decision support.

1. Introduction

Today, air pollution has become a major problem that can cause serious health problems, especially in urban areas (Usmani et al., 2020). Globally, increasing population and urbanization lead to increase the effects of air pollution. According to the United Nations estimates, the world population reached 8 billion by 2023, and is expected to reach approximately 8.5 billion by 2030 and 9.7 billion by 2050 (Gerland et al., 2022). Rapidly growing cities to meet the basic needs of the growing population have encouraged industrialization, which has led to an increase in the amount of pollutants in the atmosphere. The amount of these pollutants has increased especially in industrial centers, dense transportation networks and large residential areas. This situation has created various health risks from respiratory diseases to cardiovascular problems and has also disrupted the ecosystem balance and caused negative effects on the environment (Altikat, 2020a,b).

Air pollution has become an important problem in Türkiye, like the rest of the world, due to factors such as population growth, rapid urbanization and industrialization (Arslan et al., 2022). According to the BBC's World Air Pollution Report, Türkiye ranked 46th in the world in terms of air pollution, and Igdir was the city with the most polluted air in Europe (BBC, 2022). Igdir province, which is located in the east of Türkiye and has a population of approximately 203,000, has a critical importance in terms of air pollution. Located at the intersection of Türkiye's borders with Armenia, Azerbaijan and Iran, Igdir is prone to air pollution accumulation due to its geographical features. Igdir is located on a plateau approximately 850 m above sea level and is surrounded by high mountains. Due to this surrounded topographic structure, pollutants accumulate in the atmosphere and make the temperature inversion phenomenon even more pronounced in the winter months. This situation makes it difficult to disperse polluted

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air masses in Igdir and exacerbating the air pollution problem. Gürçam (2022) and Argun et al. (2019).

Igdir, which has a semi-arid climate, is a region where snowfall occurs frequently in the winter months and low temperatures are observed. The temperature inversion resulting from these effects causes pollutant particles, such as particulate matter 10 (PM10), particulate matter 2.5 (PM2.5), nitrogen oxides (NOx), nitrogen monoxide (NO), sulfur dioxide (SO2), and ozone (O3), to remain in the atmosphere for a long time. The temperature inversion that occurs as a result of these effects causes pollutants such as PM10, PM2.5, NO, NOx, SO2 and O3 to remain in the atmosphere for a long time. Additionally, agricultural areas and industrial facilities around Igdir also increase the amount of pollutants emitted into the atmosphere, which further worsens air quality (Altıkat, 2020; Sahin et al., 2020).

In recent years, advanced data analysis methods such as artificial intelligence and machine learning have found extensive use in studies on air pollution. In regions prone to air pollution such as Igdir, these methods are gaining importance in estimating pollutant levels by analyzing environmental and meteorological data. Deep learning algorithms based on artificial neural networks (ANNs) and long short-term memory (LSTM) networks have the capacity to predict changes in air pollutants over time with high accuracy rates. In this way, it becomes possible to predict air pollution levels in advance and develop pollution control strategies (Bekkar et al., 2021; Shakya et al., 2023; Kumar and Pande, 2023).

In this study, a comprehensive dataset was created to predict the levels of PM2.5 air pollutant that causes air pollution in Igdir province. The dataset consists of pollutant concentrations obtained from the National Air Quality Monitoring Network of the Ministry of Environment, Urbanization and Climate Change of the Republic of Türkiye (UHKIA, 2023), meteorological data provided by the General Directorate of Meteorology (MGM, 2023) and additional parameters such as solar radiation, humidity and pressure provided by NASA (NASA, 2023). In addition, time-related variables were added to the dataset in order to be suitable for both time series analysis and to examine hourly, daily and seasonal changes. In this context, the dataset was prepared in a format suitable for deep learning algorithms such as LSTM and GRU. In this process, data with different units were scaled and normalization operations were performed to increase the performance of the models.

The main purpose of this research is to evaluate the seasonal change of PM2.5 air pollutant in Igdir and its effect on air pollution by creating a comprehensive dataset. Also, it aims to make significant contributions to the prediction of air quality and understanding the effects of polluting factors in Igdir province by providing detailed analysis of environmental factors. The findings obtained are intended to support the fight against air pollution by guiding the decision-making processes of local governments and the shaping of environmental policies.

2. Related works

Studies in the literature have used traditional statistical methods such as machine learning, deep learning or hybrid methods to improve the estimation of PM2.5 like particulate matter concentrations. These studies aim to increase the accuracy of the estimation, detect changes in air pollution levels and develop solutions for pollution management. Analyses mostly conducted on hourly and daily data have evaluated the effects of meteorological factors, traffic density and industrial activities on pollution. In addition, these studies have revealed that seasonal, geographical and meteorological variables play an important role in estimation models. In this context, studies from Igdir province and other regions around the world were examined.

Batur and Babii (2024) In their study, Levenberg–Marquardt, Bayesian Regularization and Scaled Conjugate Gradient algorithms were used to estimate PM10 concentrations in Igdir Province. In the analysis made with daily data belonging to the three-year period, Bayesian Regularization algorithm exhibited the best performance with RMSE = 12.54 result. The study shows that long-term predictions considering seasonality and autocorrelation can be made in large data sets. A similar study estimated daily PM10 concentrations in Igdir using a four-layer feedforward artificial neural network (FFNN) and obtained R^2 = 0.881 and RMSE = 27.89 (Şahin et al., 2020). Studies have indicated that pollution levels are higher in winter months and seasonal effects play an important role in the models.

Tırınk and Öztürk (2023) made daily PM10 estimates for Igdir with MARS and XGBoost algorithms, XGBoost showed the best performance with R^2 = 0.833 and RMSE = 35.575. In addition, Sahin et al. (2020) analyzed air pollution problems in Igdir using Geographic Information Systems (GIS) and Analytical Hierarchy Process (AHP) methods and revealed the effects of topography and meteorological factors. In this study, it was emphasized that especially low wind speed and temperature inversion affect the pollution distribution. In the study, it was stated that the worst place in Igdir in terms of air pollution is the central district.

When looking at other studies in the world, in the study conducted by Garg and Jindal (2021), air quality was estimated with ARIMA, FBProphet, LSTM and CNN models at 12 stations in Beijing, and it was stated that LSTM stood out with MAE and RMSE values of 13.2 and 20.8 respectively. Similarly, in the study conducted by Xayasouk et al. (2020), PM2.5 was estimated in Seoul using LSTM and Denoising AutoEncoders (DAE), and it was shown that LSTM gave more accurate results. In these studies, it was observed that meteorological data and past pollution levels increased the accuracy of the estimate.

In some studies, hybrid models come to the fore. In the (Zhao et al., 2024) study, daily PM2.5 data from Dalian and Beijing cities in China between 2014–2022 were analyzed with various decomposition methods. RNN and LSTM-based hybrid models were developed and the performance of these models was compared with single models. Among these hybrid models, the HIG-LSTM model showed the best performance with the lowest error rate (R^2 = 0.88, RMSE = 13.2) in PM2.5 predictions. It was observed that the use of the hybrid model increased the prediction accuracy by effectively decomposing seasonality and data noise. In the (Shakya et al., 2023) study in India, a GRU-based encoder–decoder (GRU-ED) method was developed to predict air pollution in the city of New Delhi. This model was trained using meteorological data, vehicle count data, and emission data for hourly, 8-hourly, and daily PM2.5 forecasts. The GRU-ED model was provided better results than the other models such as Random Forest (RF) and XGBoost. As a result of model evaluation, the best performance was obtained with the GRU-ED model as R^2 = 0.959 and MAE = 1.77 in hourly forecasts.

In the study conducted by Bekkar et al. (2021), hourly PM2.5 concentrations in Beijing, China were estimated using meteorological data and PM2.5 measurements taken from surrounding stations. In the study, deep learning models such as CNN, LSTM, BiLSTM, GRU, and CNN-LSTM were compared. The CNN-LSTM model outperformed other models with the lowest error rates (MAE = 6.742, RMSE = 12.921, R^2 = 0.989). The study shows that deep learning models provide high accuracy in air pollution prediction by effectively learning both spatial and temporal features. In the use of hybrid models, Zaini et al. (2022) and Chen et al. (2020) studies are also noteworthy. In the (Zaini et al., 2022) study, PM2.5 time series in Malaysia were decomposed with the Ensemble Empirical Mode Decomposition (EEMD) method and the decomposed sub-series were predicted with LSTM. The EEMD-LSTM model was the model that gave the best results with R^2 = 0.978 and RMSE = 4.21. In the (Chen et al., 2020) study, Combinational Hammerstein Recurrent Neural Network (CHRNN) model was proposed for hourly prediction in Taiwan. CHRNN provided similar accuracy predictions at a lower cost compared to deep learning models and showed superior prediction performance in hourly PM2.5 concentrations. All of the models above have been successful in modeling long-term fluctuations and short-term sudden changes.

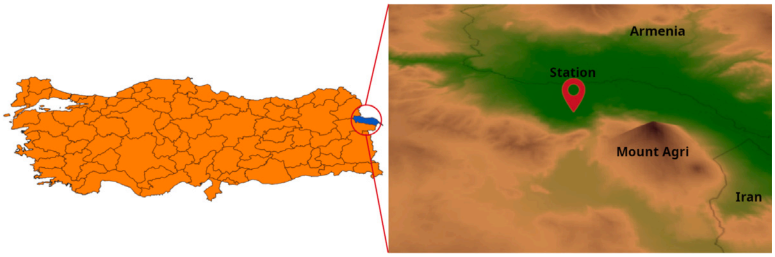


Fig. 1. Location of Igdir and station where the data is collected.

In addition to using deep learning models in air pollution prediction, they have also been used to complete missing data in data sets during the data collection process. In the (Flores et al., 2024) study, a hybrid method based on deep learning and interpolation was proposed to complete missing data in hourly PM2.5 time series in Ilo, Peru. This proposed method was compared with models such as ARIMA, LSTM, BiLSTM, GRU and BiGRU and provided higher accuracy between %2.43 and %19.96 in R^2 values. It was suggested as a suitable alternative to effectively fill short-term gaps in data sets.

As a result, when referring to the studies in the literature, it is seen that the methods and models used in air quality prediction are constantly evolving. Studies carried out especially in Igdir reveal the effect of the region's unique geographical and meteorological conditions on air pollution. Studies conducted worldwide show that deep learning and hybrid models provide more successful results than traditional methods. Deep learning models like LSTM, GRU, CNN and their hybrid versions provide high accuracy predictions by effectively learning both spatial and temporal features. In addition, deep learning models have been shown to be effective in data preprocessing stages such as completing missing data.

3. Materials and methods

3.1. Data

The dataset used in this study was created to estimate air pollution levels in Igdir province and includes data obtained from three different sources. PM10, PM2.5, SO2, NOX, NO, NO2 and O3 air pollutants obtained from the National Air Quality Monitoring Network of the Ministry of Environment and Urbanization of the Republic of Türkiye (UHKIA, 2023), temperature, wind speed, wind direction, total precipitation and relative humidity data obtained from the General Directorate of Meteorology of the Ministry of Environment and Urbanization of the Republic of Türkiye (MGM, 2023), and surface pressure, relative humidity at 2 m, all-sky distributed irradiance and all-sky ultraviolet index data obtained from NASA's Prediction Of Worldwide Energy Resources (NASA, 2023) platform were included in the dataset. The dataset consisting of hourly measurements was organized in a time series format and structured to allow the analysis of daily, weekly and seasonal trends. With the addition of year, month, day and hour information, the total number of parameters in the dataset reached 20. The location of Igdir province within Türkiye and the location of the station where the data was collected are shown in Fig. 1.

3.2. Feature selection, missing values and imputation

The dataset created for this study consists of 27,019 hourly data records between 17.11.2020 and 18.12.2023. Data outside this date range were removed from the data set because they contained a large number of consecutive and missing records. When the records in the selected date range were examined, it was observed that there were missing records in various parameters. In this dataset, missing data analysis was performed to determine the distribution and areas where the missing data were concentrated. A missing data heat map for variables with missing data and a table showing the number of missing data are given in Fig. 2.

As a result of the missing data analysis, it was observed that there was a maximum of %6.7 missing in some variables. While the missing data in air pollution parameters showed an irregular distribution, the missing data in meteorological parameters were concentrated at certain times. It is thought that these missing data are due to reasons such as measurement errors in the data collection process or sensors not working at certain periods. Since the low rate of missing data would not pose a major problem in terms of the overall analysis of the data set, it was decided to complete these missing data.

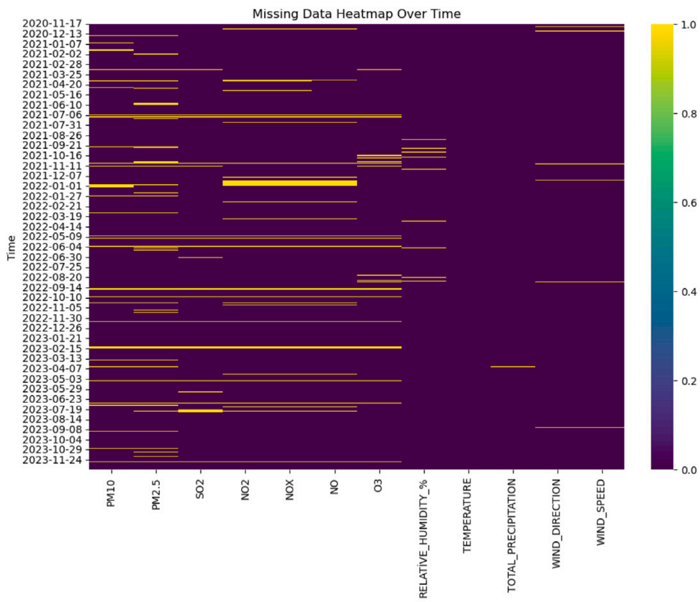
Before completing the missing data, a correlation matrix is created to evaluate the relationship between the parameters in the data set. Pearson correlation coefficient describes the relationship between two variables and is one of the most commonly used methods to measure the linear relationship between two variables (Cohen et al., 2009). Pearson correlation coefficient (r) is calculated as shown in Eq. (1):

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i-\bar{x})^2}\sqrt{\sum(y_i-\bar{y})^2}} \tag{1}$$

The correlation matrix provides important information for the analysis of the data set by numerically revealing the relationship between the variables in the data set. When the correlation matrix given in Fig. 3 is examined, it is observed that there are strong positive correlations between air quality parameters, especially "PM10" and "PM2.5". High positive correlations between air pollutants such as PM10, PM2.5, SO2, NO2, NOX show that these pollutants generally increase and decrease together. There are generally weak or moderate correlations between air pollutants and time information (Year, Month, Day, Hour). This shows that the change in air pollution over time is complex and cannot be explained only by time. Similarly, it was found that there is a significant correlation relationship between the most commonly measured meteorological parameters such as temperature, humidity, wind speed, precipitation, and air pollution. While temperature and wind speed in particular show a negative correlation with air pollutants other than O3, humidity has a positive relationship with particulate pollutants; This shows that particle density increases in humid weather. There is almost no correlation between the amount of precipitation and air pollutants.

Considering the low rate of missing data, the presence of strong correlations between some variables, and the variability of the data over time, more than one method was used for filling missing data. In data with time series features such as humidity, temperature, and wind speed, missing data were generally filled with linear interpolation between the previous and next values. Since precipitation data was mostly zero, the missing data for this parameter was filled with zero. Because the parameter wind direction is a categorical variable, missing data for this parameter was filled with the previous or next data.

Then, the missing data for PM10, PM2.5, NO2, NO, NOX, and O3 were completed with interpolation for missing values less than 6



Variable	Missing Count
PM10	1323
PM2.5	1801
SO2	1145
NO2	1656
NOX	1675
NO	1631
O3	1176
RELATİVE_HUMIDITY_%	478
TEMPERATURE	9
TOTAL_PRECIPITATION	50
WIND_DIRECTION	559
WIND_SPEED	559

Fig. 2. Missing data heatmap and counts.

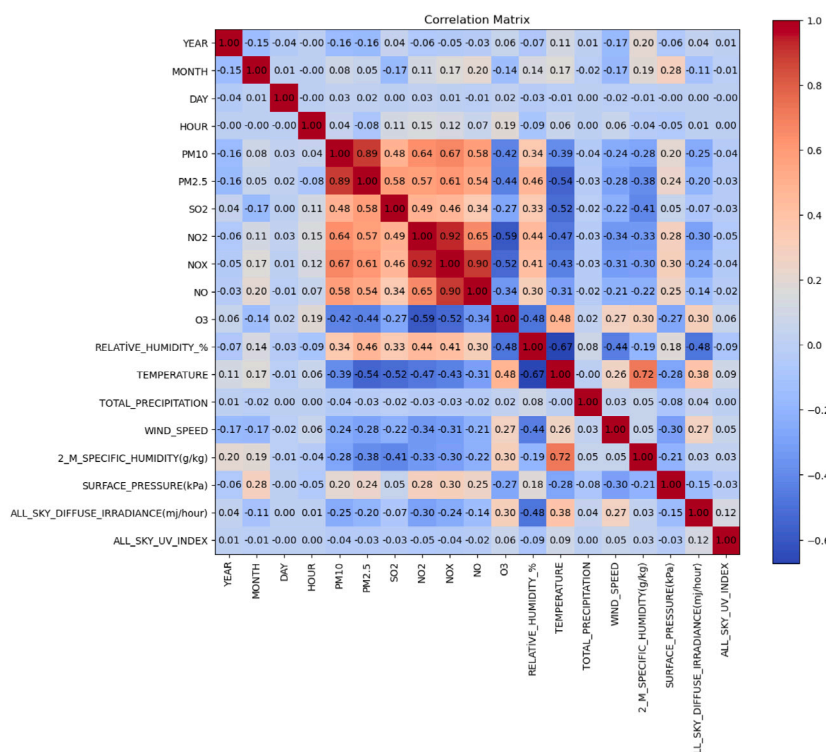


Fig. 3. Correlation matrix for unfilled data.

consecutive values in order not to disrupt seasonal trends, and the parts with more missing values were filled using the data of the previous or next day. After the missing data filling process was completed, the correlation matrix of the first data set containing the missing data was compared with the completed data set. Once the results were examined, it was observed that there were very small changes in the correlation values, and this was interpreted as the relationships of the parameters in the data set were preserved, therefore the obtained data set was reliable for analysis. The graph of the correlation differences is also indicated in Fig. 4.

3.3. Encoding categorical variables and cyclical encoding of temporal features

The effect of wind on air quality has long been known. It is widely known that low wind speeds promote the formation of smoggy air events, especially in cities. Not only the speed of the wind, but also its direction plays a significant role in air pollution levels. In cities, it has been found that certain wind directions directly affect air quality by carrying or depositing pollutants. For this reason, it is important to examine the effects of wind on air pollution in detail with the factors of direction and speed. As seen in the literature, low wind speeds

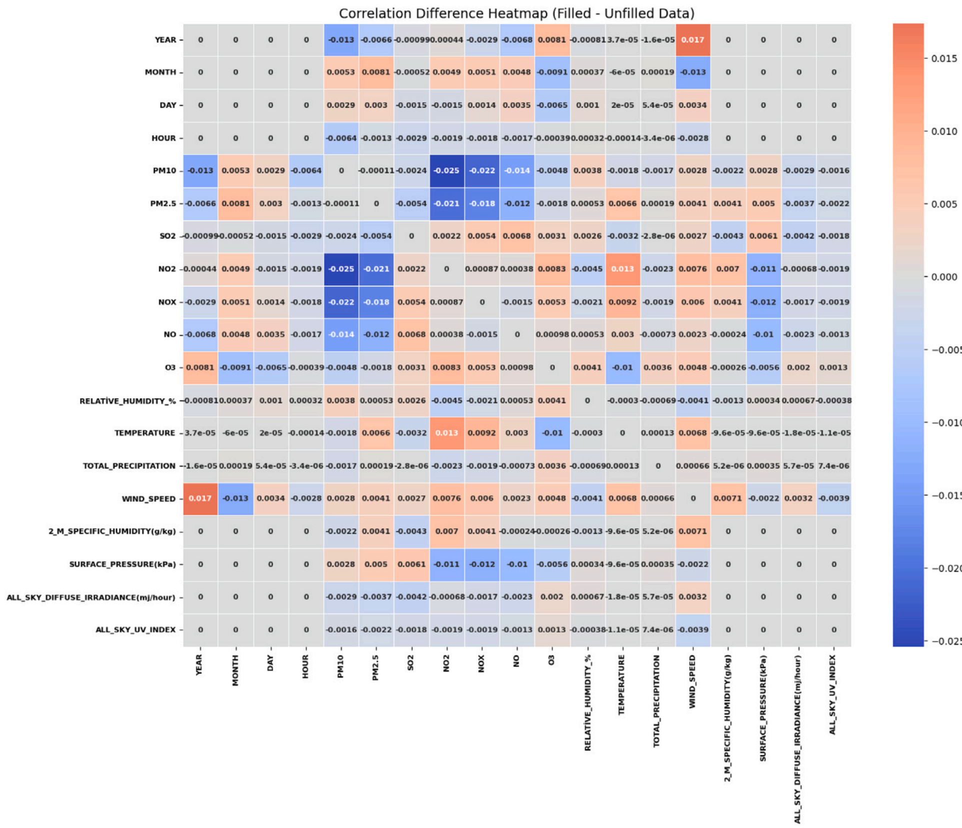


Fig. 4. Correlation difference heatmap.

cause pollutants to remain in the atmosphere for a longer time and accumulate, while the wind direction determines the spreading area of the pollutants (Bekkar et al., 2021).

In this study, the accuracy of air pollution analysis was increased by converting the wind direction variable in the data set into a numerical format. The wind direction in the data set was categorically represented by 16 different directions, namely N, NNE, NE, ENE, E, ESE, SE, SSE, S, SSW, SW, WSW, W, WNW, NW and NNW, as well as the "C" category, which represents the cases where there is no wind. In order to make this categorical data suitable for the numerical model, an angle was assigned to each direction, increasing clockwise. The north direction was initially determined with the value of 0°, and each direction was separated by 22.5° angles on the compass. Thus, the wind direction was represented by angle values ranging from 0° to 337.5° degrees. Wind directions and the angle values corresponding to these directions are shown in Fig. 5.

As shown in Fig. 5, the angle values of the wind directions were converted to sine and cosine components in order to support the numerical model. The values of the sine and cosine components were added to the data set as two new columns. In this way, the direction data was expressed as a vector on a two-dimensional plane (x-y axis). Since category C shows the situations where there is no wind, the sine and cosine values corresponding to this category were assigned as zero. With this method, it was aimed to analyze the accumulation and dispersion effects of air pollutants depending on the direction and speed of the wind more precisely.

Additionally, temporal variables such as month, day, and hour inherently exhibit cyclical characteristics. Even though this data is sequential in the data set, it may cause the models to not be able to understand the temporal features sufficiently because it is used in training the models in its current form. To address this issue, a similar approach to the transformation applied to wind direction was adopted. Specifically, sine and cosine transformations were applied to the hour, day, and month variables. As a result, two new features were created for each temporal variable and added to the dataset. This method is widely used in the literature and has been applied in various studies to improve model performance (Zhou et al., 2021; Ma et al., 2024). Through this transformation, the model can better understand relationships such as the proximity between hour 23 and hour 0, or between January (month 1) and December (month 12), which are otherwise distant in their raw numerical forms.

3.4. Detection and removing outlier data

In time series analysis, detecting and processing outliers is an important step to increase the data set accuracy and the model performance. In this study, outliers for PM2.5 measurements were detected with a dynamic method using daily sliding averages and standard deviations. Firstly, the sliding mean and standard deviation values were calculated using a 24-h (1-day) time window on the PM2.5 values in the dataset. Then, the following Eqs. (2) and (3) were used to determine the boundaries of the outliers:

$$upperbound = \mu + 2\sigma \tag{2}$$

$$lowerbound = \mu - 2\sigma \tag{3}$$

Values outside these limits were defined as outliers. Detected outliers were replaced with an upper bound for values exceeding the upper bound, and with a lower bound for values below the lower bound. Thus, the effect of outliers in the data set was reduced and the data set used in the analysis was made more consistent. As a result of this process, a total of 668 outliers in the PM2.5 data were detected and corrected. This approach is one of the standard deviation-based methods widely used in the literature and allows the determination of dynamic limits suitable for trends, especially in time series data.

Code	Angle	Sin	Cos
C	-	0	0
N	0°	0	1
NNE	22.5°	0.38	0.92
NE	45°	0.7	0.7
ENE	67.5°	0.92	0.38
E	90°	1	0
ESE	112.5°	0.92	-0.38
SE	135°	0.7	-0.7
SSE	157.5°	0.38	-0.92
S	180°	0	-1
SSW	202.5°	-0.38	-0.92
SW	225°	-0.7	-0.7
WSW	247.5°	-0.92	-0.38
W	270°	-1	0
WNW	292.5°	-0.92	0.38
NW	315°	-0.7	0.7
NNW	337.5°	-0,38	0.92

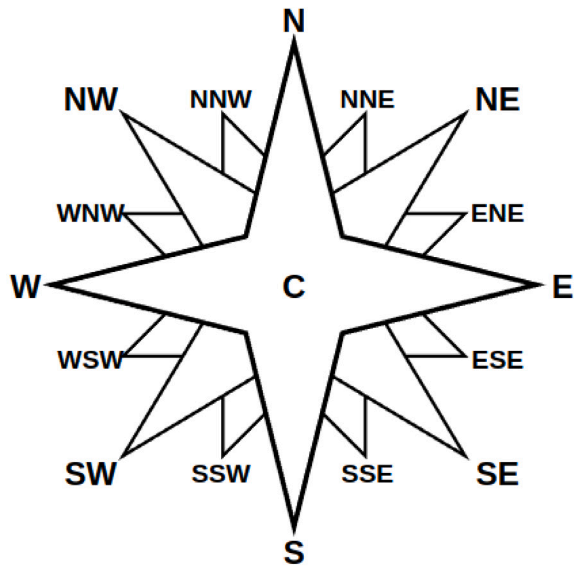


Fig. 5. Wind directions encoding.

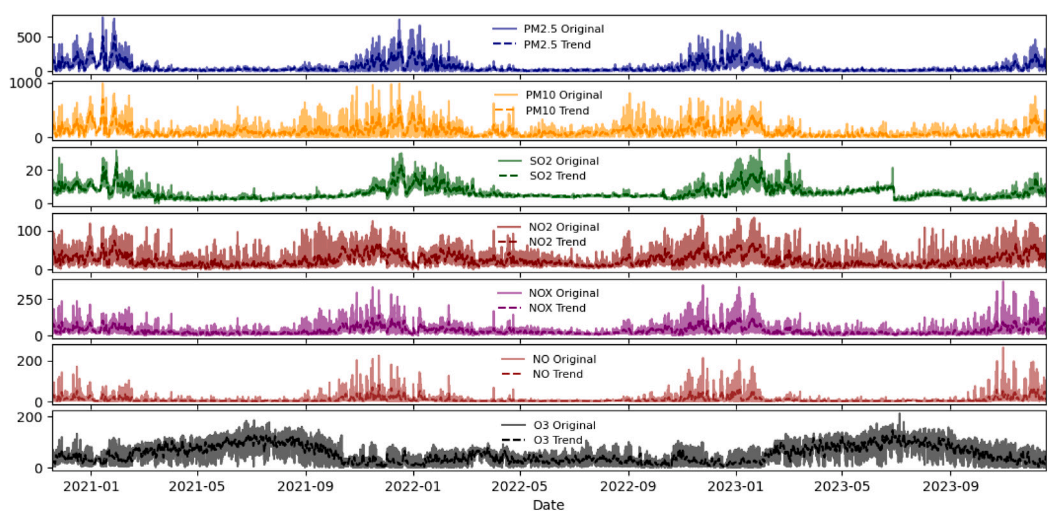


Fig. 6. Temporal variation of air pollution parameters.

3.5. Normalization

In order to increase the accuracy of the models to be trained, the values of the parameters in our data set were normalized with min–max normalization technique. This method is used to scale each value between 0 and 1. Min–max normalization was applied according to Eq. (4) by taking into account the minimum and maximum values of each feature in the data set:

$$x = \frac{x - min}{max - min} \tag{4}$$

This process ensures that the features are in the same scale range and thus helps the model to benefit equally from features with different value ranges. In this way, the prediction accuracy can be improved. Ali et al. (2014)

3.6. Temporal behavior of variables

Time series data of each key variable was visualized to understand the temporal changes of the variables in the data set. These visualizations aim to provide information about trends, seasonal changes and possible anomalies. Since the PM2.5 air pollution parameter is determined as the predictor variable in this study, the relationships with PM2.5 are primarily evaluated while analyzing the behavior of other variables in the dataset over time. In particular, understanding the time-dependent change of PM2.5 may be beneficial to improve the accuracy of forecast models. For this reason, time series graphs of the variables in the dataset are included in this section.

When Fig. 6 is examined, significant seasonal and periodic fluctuations are observed for the PM2.5 parameter. Particularly in the winter months, significant increases are observed in PM2.5 levels. These increases can be attributed to factors such as heating activities, low temperatures and limited air circulation. PM10 also exhibits a similar behavior. Furthermore, there is a strong correlation between PM10 and PM2.5 parameters. It is observed that there is a slight decrease in the concentrations of PM2.5 and PM10 parameters in the winter months, although not very significant compared to previous years. Similar to these two parameters, fluctuations depending on time were also observed clearly in gas pollutants such as SO2, NO2, NOX and NO. The peaks of NO and NOX pollutants over time generally overlap. SO2, on the other hand, follows a relatively constant course, but exhibits

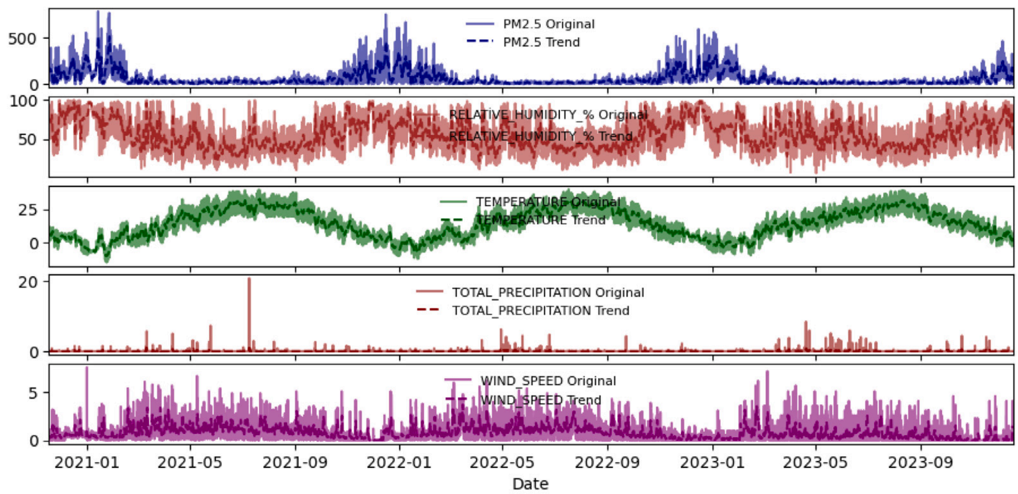


Fig. 7. Temporal Variation of Meteorological Features.

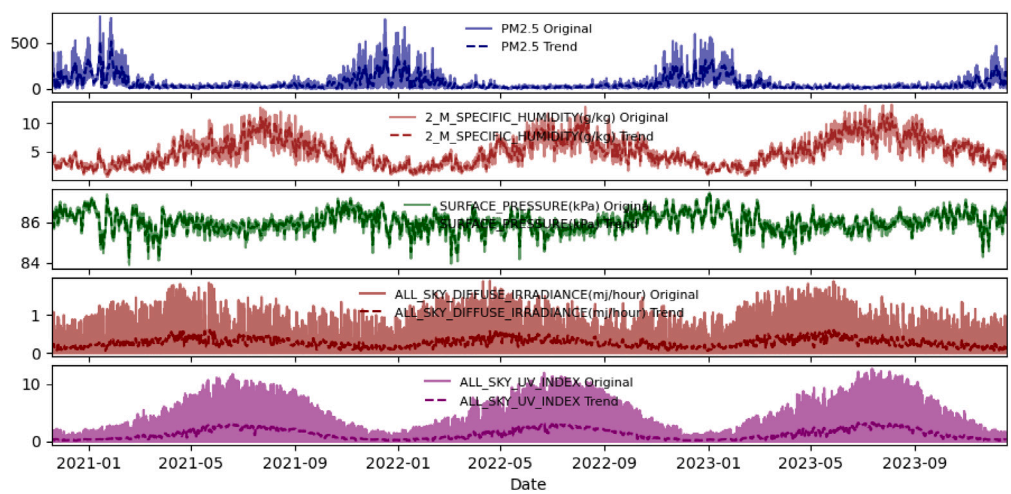


Fig. 8. Temporal variation of meteorological parameters from NASA POWER.

some sudden increases thought to be related to industrial activities or energy production. Ozone (O3) levels follow a different pattern from other pollutants, increasing especially in the summer months.

When the relative humidity, one of the meteorological parameters given in Fig. 7, is examined, it follows a relatively constant course over time, while it shows significant increase in the winter months. The temperature graph clearly reflects the seasonal cycles, peaking in the summer months and decreasing in the winter months. While the temperature parameter exhibits an inversely proportional relationship with most pollutants, including PM2.5, it has been observed that this relationship is positive with the ozone parameter. The total precipitation often shows sudden peaks and exhibits an irregular distribution. It is known that such precipitation events cause a washing effect that temporarily reduces the concentration of particulate matter. Wind speed, on the other hand, varies throughout the year, and it is thought that high speeds reduce pollution levels and increase distribution.

Fig. 8 shows the relationship between PM2.5 and other meteorological parameters provided by NASA (NASA, 2023). The specific humidity parameter at 2 m increases in the summer months and has a negative relationship with PM2.5. The surface pressure parameter is relatively stable throughout the year, while its correlation with PM2.5 is low. All sky dispersed irradiance and all sky UV index, as variables associated with sunlight, increase in summer and exhibit a negative relationship with PM2.5.

3.7. Deep learning models

3.7.1. LSTM and Bi-LSTM networks

Long Short-Term Memory (LSTM) is a recurrent deep learning network architecture proposed by Hochreiter and Schmidhuber to eliminate the disadvantages of traditional RNN algorithms (Hochreiter and Schmidhuber, 1997). Compared to RNN, it can overcome both long-term dependency problems and reduce the probability of gradient disappearance (Chen et al., 2019). It basically consists of cell structure (cell state) and control mechanisms called gates. Cell state stores long-term information. The gates decide what information is taken into the cell, what information is forgotten, and what information is transferred as output. These gates are the forgetting gate, entry gate and exit gate, respectively (Greff et al., 2016). LSTM structure is shown in Fig. 9.

Mathematically, these mechanisms of LSTM use sigmoid and hyperbolic tangent functions. One of the most important components of LSTM, the forget gate, determines which information in the cell will be deleted. The output of the forget gate (f_t) is calculated with the sigmoid activation function given in Eq. (5). It is obtained by combining the previous hidden state (h_{t-1}) with the current input (x_t).

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \tag{5}$$

In Eq. (5), x_t represents the current input. h_{t-1} represents the value of the latent state in the prior time step. b_f represents the bias value,

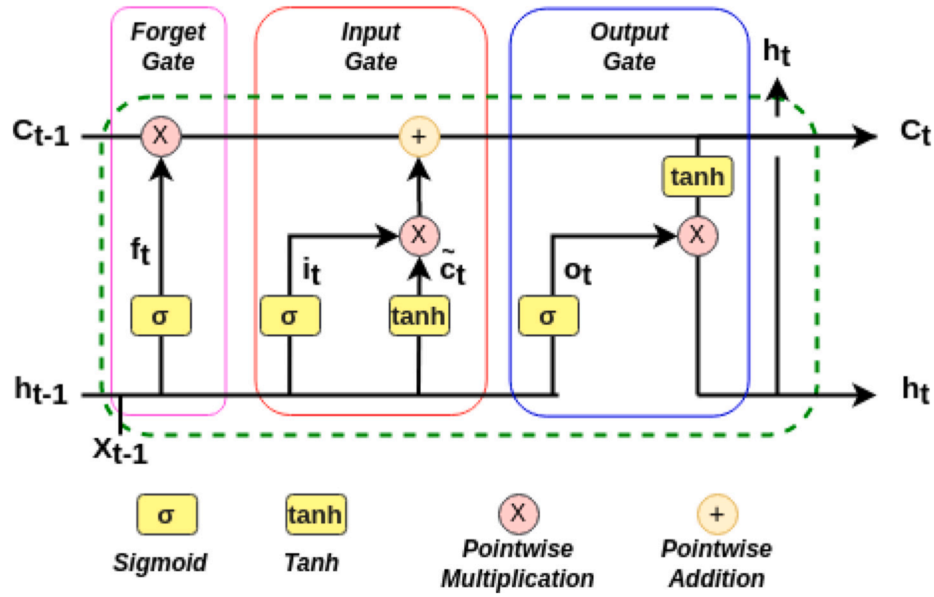


Fig. 9. LSTM architecture.

while W_f represents the weights. According to the value of f_t, the information from the previous cell is either kept or forgotten. Another critical mechanism of LSTM is the input gate. This gate controls the new information to be added to the cell. This process takes place in two stages: First, the sigmoid (Eq. (6)) function is used to decide which information will be added. Then, the candidate values of the new information are created using the hyperbolic tangent function (Eq. (7)). Then, the information to be added to the cell state is produced by multiplying the obtained i_t and $\tilde{c}_t$ values. After this step, the cell state (c_t) is updated. As shown in Fig. 9, the new cell state is obtained by adding the information from the forget gate ($f_t.c_{t-1}$) and the information from the input gate ($i_t.\tilde{c}_t$) (Eq. (8)).

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \tag{6}$$

$$\tilde{c}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \tag{7}$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \tag{8}$$

The last gate in the LSTM cell is the output gate. The output gate determines which of the information stored in the cell will be used as output. The sigmoid function selects which information will be transferred (Eq. (9)), while the cell state is scaled with the tanh function. Finally, by combining these two information (Eq. (10)) the new hidden state (h_t) is produced.

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \tag{9}$$

$$h_t = o_t \odot \tanh(c_t) \tag{10}$$

Besides the LSTM network, there are also BiLSTM networks known as bi-directional LSTM. Bidirectional architectures refer to an approach that aims to learn both past and future information. One of the first examples, bidirectional recurrent neural networks (BiRNN), was introduced by Schuster and Paliwal (1997). Bi-LSTM networks were created using a bidirectional structure in LSTM networks. In LSTM networks, cell state information is transmitted unidirectionally from beginning to end. For this reason, LSTM networks only learn information from past time steps. In order to overcome this limitation, two separate cell state conveyor belts are used in BiLSTM networks. In this way, BiLSTM learns information from both prior and posterior data.

In a Bi-LSTM network, two separate LSTM cells work simultaneously on the same data. One of these LSTM cells (forward LSTM) handles the data forward in time, while the other (backward LSTM) handles the data in the reverse direction. At each time step, the hidden states h_t^{forward} and h_t^{backward} obtained from forward and backward LSTM (Eqs. (11) and (12)) are combined. Thus, the hidden output of BiLSTM (Eq. (13)) h_t is obtained. This combination makes it possible to consider both past $(t-1)$ and future $(t+1)$ contexts and increases of the learning capacity of the model (Graves et al., 2013; Bian et al., 2020).

$$h_t^{\text{forward}} = \text{LSTM}_{\text{forward}}(x_t, h_{t-1}^{\text{forward}}) \tag{11}$$

$$h_t^{\text{backward}} = \text{LSTM}_{\text{backward}}(x_t, h_{t+1}^{\text{backward}}) \tag{12}$$

$$h_t = [h_t^{\text{forward}}; h_t^{\text{backward}}] \tag{13}$$

3.7.2. GRU and BiGRU networks

Gated Recurrent Units (GRU) were proposed by Cho et al. (2014) to enable each recurrent unit to capture dependencies at different time steps. It is a RNN architecture widely used in areas such as time series analysis, natural language processing and sequential data processing. GRU was developed specifically to solve the problems of long-term dependencies not being learned and vanishing gradients, which are frequently encountered in RNNs (Chung et al., 2014). The structure of GRU consists of two main gate mechanisms. These gates are update gate and reset gate. The update gate determines how much of the old information will be preserved and how much of the new information will be received, while the reset gate controls how much of the past state information will be forgotten (Cho et al., 2014). The structure of the GRU cell is shown in Fig. 10.

As shown in Fig. 10, the inputs and outputs to a GRU cell are the hidden vector x_t at the current time step, the hidden state h_{t-1} at the previous time step, and the output h_t at the current time step. The mathematical operations that GRU performs at each time step can be expressed as follows: The reset gate (r_t) determines how much of the previous hidden state (h_{t-1}) will be forgotten. Mathematically, the reset gate is defined as in Eq. (14). In this equation, W_r represents the learnable weight matrix of the reset gate.

$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t]) \tag{14}$$

The update gate (z_t) determines how much of the previous hidden state will be preserved and how much will be updated. Mathematically,

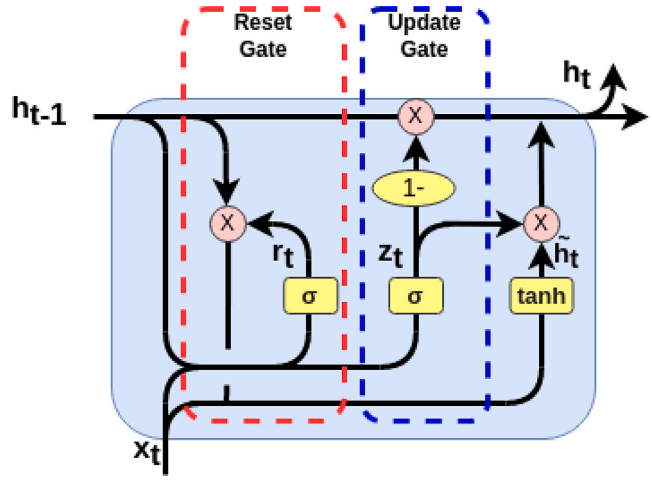


Fig. 10. GRU architecture.

it is defined as in (15). In this equation, W_z is the learnable weight matrix of the update gate.

$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t]) \tag{15}$$

Then, to calculate the current hidden state, the temporary hidden state $\tilde{h}_t$ is calculated using the output of the reset gate, r_t. *tanh*, used as the activation function, limits the values of the temporary state to the range $[-1, 1]$, allowing the model to learn more stably. The calculation formula for the temporary hidden state is given in Eq. (16). In this equation, W represents the learnable weight matrix.

$$\tilde{h}_t = \tanh(W \cdot [r_t * h_{t-1}, x_t]) \tag{16}$$

In the last step, a weighted average is taken between the previous hidden state (h_{t-1}) and the temporary hidden state ($\tilde{h}_t$) using the update gate (z_t). As a result of this process, the final hidden state h_t is calculated and mathematical formula is as shown in Eq. (17).

$$h_t = z_t * h_{t-1} + (1 - z_t) * \tilde{h}_t \tag{17}$$

Another network that uses the GRU architecture is the Bidirectional Gated Refresh Unit (BiGRU). BiGRU can be defined as GRU's bidirectional processing of data. BiGRU consists of two separate layers, namely forward GRU ($\overrightarrow{h_t}$) and backward GRU ($\overleftarrow{h_t}$). By combining the hidden states obtained from both layers, the final output (h_t) is obtained (Deng et al., 2019).

BiGRU mathematical expressions are based on equations used in the standard GRU. However, these equations have been extended to include bidirectional data flow. The calculations and final output states of both forward and backward GRU cells are expressed in Eq. (18) and Eq. (19).

For forward GRU:

$$\begin{aligned}
\overrightarrow{r_t} &= \sigma(W_r \cdot [\overrightarrow{h}_{t-1}, x_t]) \\
\overrightarrow{z_t} &= \sigma(W_z \cdot [\overrightarrow{h}_{t-1}, x_t]) \\
\overrightarrow{\tilde{h}_t} &= \tanh(W \cdot [\overrightarrow{r_t} * \overrightarrow{h}_{t-1}, x_t]) \\
\overrightarrow{h_t} &= \overrightarrow{z_t} * \overrightarrow{h}_{t-1} + (1 - \overrightarrow{z_t}) * \overrightarrow{\tilde{h}_t}
\end{aligned} \tag{18}$$

For backward GRU:

$$\begin{aligned}
\overleftarrow{r_t} &= \sigma(W_r \cdot [\overleftarrow{h}_{t+1}, x_t]) \\
\overleftarrow{z_t} &= \sigma(W_z \cdot [\overleftarrow{h}_{t+1}, x_t]) \\
\overleftarrow{\tilde{h}_t} &= \tanh(W \cdot [\overleftarrow{r_t} * \overleftarrow{h}_{t+1}, x_t]) \\
\overleftarrow{h_t} &= \overleftarrow{z_t} * \overleftarrow{h}_{t+1} + (1 - \overleftarrow{z_t}) * \overleftarrow{\tilde{h}_t}
\end{aligned} \tag{19}$$

The final output (h_t) is obtained by combining the forward GRU and backward GRU hidden states as shown in Eq. (20):

$$h_t = [\overrightarrow{h_t}; \overleftarrow{h_t}] \tag{20}$$

3.7.3. CNN

Convolutional neural networks (CNN) are widely used as deep learning models, especially in the fields of image processing and computer vision. Unlike traditional artificial neural networks (ANN), they consist of convolution layers, activation functions, pooling layers and fully connected layers. The convolution layer is the layer where local feature maps are extracted using filters (kernels) on the image. These filters are applied by sliding on the image and enable the detection of low-level features such as edges and textures (LeCun et al., 1998).

Pooling layers reduce the computational cost by reducing the size of feature maps. This layer prevents overfitting by preserving distinct features. The most commonly used methods are max pooling and average pooling (Krizhevsky et al., 2012). Fully connected layers are usually layers where operations such as regression and classification are performed.

3.8. Performance evaluation

Another issue that is as important as the quality of the dataset and the models to be trained is how to evaluate the performance of these models. Although there are multiple criteria to measure how well the model performs, the metrics to be used in this work as follows: Mean Absolute Error (MAE), Root Mean Square Error (RMSE) and Coefficient of Determination (R^2). In the formulas given for these metrics, y_i, $\hat{y}_i$, $\bar{y}$ and n denote the actual value, predicted value, average value and total number of samples, respectively.

In addition to evaluating model performance across the entire dataset, it was also assessed specifically for peak values. These peak values represent sudden and extreme increases in pollutant concentrations and were dynamically identified using the Z-score method with a threshold of 1.5. In a similar study (Hu et al., 2025), peak values were identified based on a fixed threshold derived from the local air quality index, where measurements exceeding this value were classified as peaks. However, an examination of the dataset used in this study revealed sudden fluctuations in air pollution levels, along with significant seasonal variations in average concentrations. Particularly during the summer months (Fig. 7), pollution levels were observed to be quite low, while in winter months, they increased substantially. In such cases, applying a fixed threshold may fail to detect certain peak events. Therefore, a dynamic approach was employed, using moving daily averages and standard deviations to identify peak values. This method aims to provide a more accurate peak analysis by capturing both seasonal trends and abrupt increases in pollution levels.

3.8.1. R^2

The determination coefficient indicates how much of the variation in the dependent variable can be explained by the independent variables. It takes values between 0 and 1, and the closer the R^2 value is to 1, the better the model explains the changes in the dependent variable. R^2 is calculated with the help of the following equation (Eq. (21)):

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{21}$$

3.8.2. RMSE

RMSE is defined as the root mean square of all errors. This measure is used to evaluate the size of the prediction error and the accuracy of the model. RMSE is more sensitive to large errors, so it is preferred when precision is important (Chicco et al., 2021). RMSE is calculated using the following equation (Eq. (22)):

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{22}$$

3.8.3. MAE

Mean Absolute Error (MAE) is a performance measure that calculates the average of the absolute differences between a model's predicted values and the actual values. MAE, which evaluates the magnitude of error regardless of direction, is widely used to understand the accuracy of predictions. Model predictions show us that a low MAE value is close to the true values, whereas a high MAE value, on the contrary, is farther from the true values and contains more errors. It is easy to interpret and less sensitive to large errors, as it directly expresses the magnitude of the prediction errors, (Chicco et al., 2021). The formula for calculating MAE is given in Eq. (23):

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{23}$$

4. Results and discussion

In this section, experimental results are presented and findings obtained from training and testing various deep learning models to predict PM2.5 concentrations are discussed. In addition to the Python programming language, various Python packages including Pandas, Numpy, Scikit-Learn, Keras and TensorFlow were used to process the data and create the models. The calculations required for training the models were performed at the TÜBİTAK ULAKBİM, High Performance and Grid Computing Center (TRUBA) (Tubitak, 2011).

Firstly, the air pollution dataset was separated into training (%90) and test (%10) datasets without changing the data order in order to be suitable for time series analysis. In order to understand the effect of past data on the predictions, LSTM, GRU, BiLSTM, BiGRU, CNN-LSTM and CNN-GRU artificial neural networks were created using 8, 24 and 72 h time windows. The best models were tried to be created for different time windows using different hyperparameters. The models were planned to be trained for a total of 100 epochs, but an Early Stopping was used to avoid overfitting and eliminate unnecessary computational load. In the early stopping mechanism, the patience value was set to 10. This means that the training process will be stopped when the loss value in the validation dataset does not improve for 10 consecutive epochs. The loss function used in the training process was the Mean Squared Error (MSE).

While LSTM, GRU, BiLSTM and BiGRU of the trained artificial neural networks are completely recurrent neural network (RNN)-based structures, 1D convolutional (CNN) layers are used in the CNN-LSTM and CNN-GRU models to capture spatial features in the input before learning temporal dependencies. Therefore, the hyperparameter set used in these models is slightly different. In all models, Batch size, Optimizer and Dropout common parameter sets were used. Batch size values were selected as 16, 32 and 64, while Adam and RMSprop were used for Optimizer. Dropout rates were selected as 0.1 and 0.2.

In LSTM, GRU, BiLSTM and BiGRU models, 50, 100, 150 and 200 neurons were used as hyper parameter sets to test different neuron numbers. Similarly, 2, 3 and 4 were used for the hidden layer numbers in these models. The proposed structure for these 4 models is also shown in Fig. 11.

In the CNN-LSTM and CNN-GRU models, the dataset is first passed through the CNN layers and the resulting outputs are provided as input to the LSTM and GRU stages. The hyperparameters in these two models differ due to this structure. First, the hidden layer number hyperparameters for the CNN layers to be applied were determined as 1 and 2. The hyperparameter set determining the number of neurons in the CNN layer was selected as 64 and 128. The hidden layer number parameters for the layers outside the CNN layer were determined as 2 and 3, and the number of neurons in these layers was determined as 100 and 200. The proposed structure for the CNN-LSTM and CNN-GRU models is shown in Fig. 12.

Architecture and hyperparameter sets of the models were determined and the models were trained in the next step. For each of the LSTM, GRU, BiLSTM, and BiGRU artificial neural networks, 432 different models were trained, while for each of the CNN-LSTM and CNN-GRU artificial neural networks, 576 different models were trained. The results of these models were examined and ranked according to the root mean square error ($RMSE$) and determination coefficient (R^2). Then, they were grouped according to time windows. The hyperparameters of the models giving the best results are given in Tables 1 and 2. The R^2, $RMSE$ and MAE values of these models are given in Table 3.

Table 1
Best hyperparameters for LSTM, GRU, BiLSTM and BiGRU with different time windows.

Lag	Model	Optimizer	Batch size	Hidden layers	Neurons	Dropout
8 h	LSTM	Adam	16	2	100	0.2
	BiLSTM	Adam	32	2	150	0.2
	GRU	Adam	16	4	100	0.1
	BiGRU	RMSprop	16	3	100	0.1
24 h	LSTM	Adam	32	2	50	0.2
	BiLSTM	Adam	16	4	100	0.2
	GRU	Adam	64	4	200	0.1
	BiGRU	Adam	16	3	200	0.1
72 h	LSTM	Adam	16	2	200	0.1
	BiLSTM	Adam	16	3	150	0.1
	GRU	RMSprop	16	2	100	0.1
	BiGRU	Adam	64	4	200	0.2

Beyond the model architecture, effectiveness of deep learning models in time series forecasting requires careful consideration of the hyperparameters used and the forecast range (time window). In this study, the performances of LSTM, GRU, BiLSTM, BiGRU, CNN-LSTM and CNN-GRU models were examined in 8, 24 and 72-h time windows and the main factors affecting this performance were analyzed. The findings show that the model performance largely depends on the time window and hyperparameter settings. Especially for the 8 and 24 h windows, models achieved lower error values. The LSTM model reached the best predictive performance for the 24 h window with MAE = 9.65 and R^2 = 0.949, while the GRU model showed strong performance in the 8-h window with MAE = 9.93 and R^2 = 0.944. For the 72 h window, the BiLSTM model outperformed the others. In general, all models exhibited significantly improved accuracy in the 24 h window, whereas a decline in performance was observed in the 72 h window. Notably, GRU and LSTM architectures demonstrated stable performance across different time windows.

In terms of optimization algorithms, the Adam optimizer generally yielded the best results, while models trained with RMSprop performed relatively poorly. Interestingly, the BiLSTM model, despite its complexity, showed lower accuracy than expected, indicating that increased model complexity does not necessarily translate to improved performance. It was observed that tuning the number of neurons and hidden layers played a critical role in model success.

In addition to evaluating general performance metrics, assessing model performance on peak (extreme) values is also crucial, particularly in urban areas where high levels of air pollution are common. In this context, the CNN-LSTM model stood out for peak prediction, achieving RMSE = 28.16, R^2 = 0.792, and MAE = 22.45 in the 8-h window. The LSTM model showed a balanced performance in the 24-h time window, with good results for both general predictions (MAE: 9.65, RMSE: 14.36) and peak value predictions (RMSE = 29.30, R^2 = 0.774). On the other hand, BiLSTM and BiGRU models, which have more complex structures, performed worse in predicting peak values, especially in the 72-h window (e.g., BiLSTM RMSE = 33.95, R^2 = 0.683). This indicates that more complex models do not always lead to better results for predicting extreme values.

Overall, the models showed strong performance, with R^2 values ranging from 0.935 to 0.949 and R^2 values for peak values ranging from 0.659 to 0.792. These results indicate that the models can explain

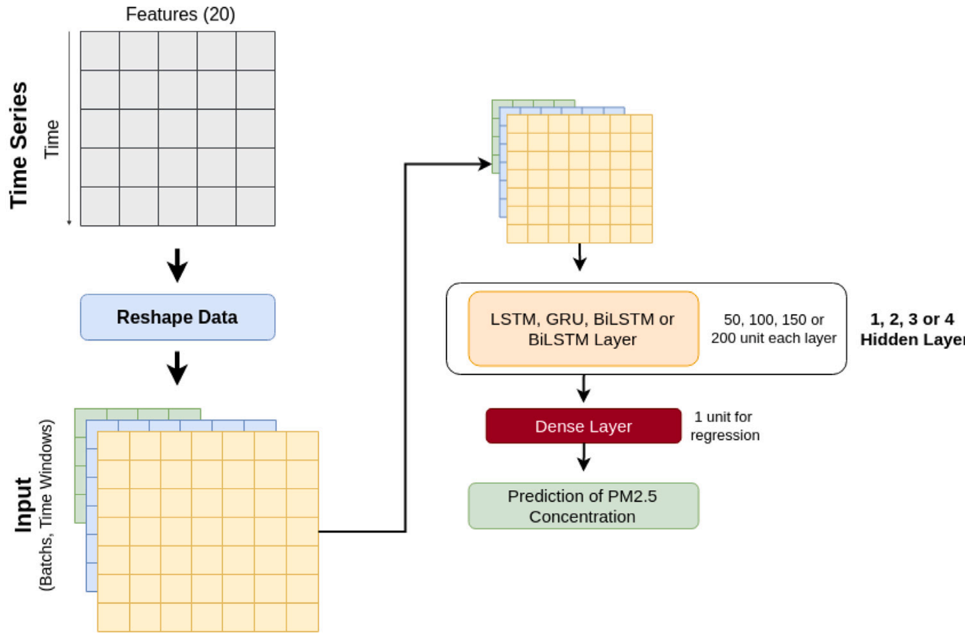


Fig. 11. Proposed LSTM, GRU, BiLSTM and BiGRU models.

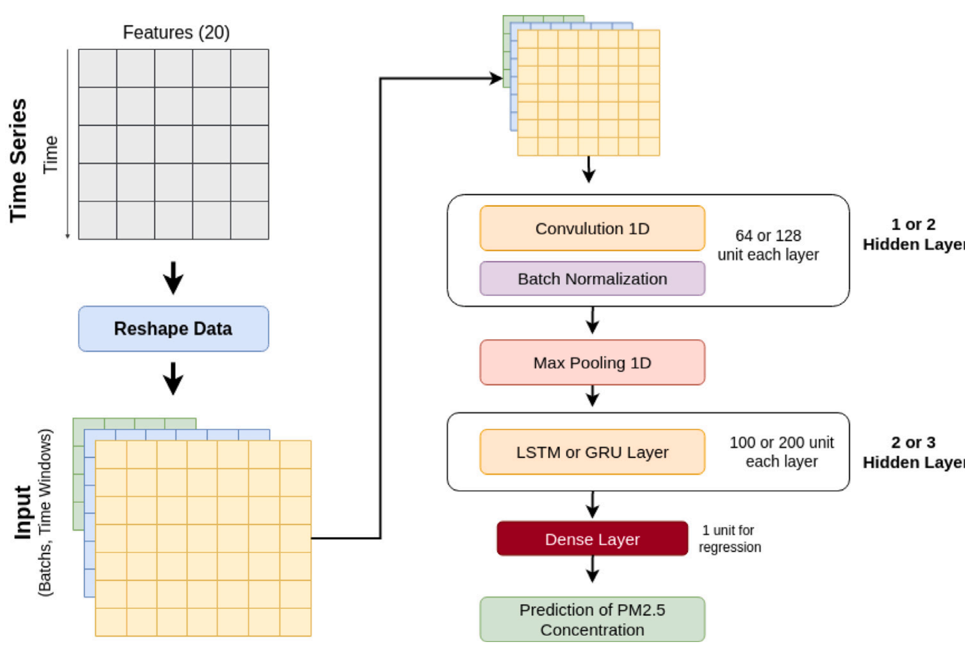


Fig. 12. Proposed CNN-LSTM and CNN-GRU models.

Table 2
CNN-LSTM and CNN-GRU models best hyperparameters for different time windows.

Lag	Model	Optimizer	Batch size	Conv. layers	Conv. neurons	Layers		Neurons		Dropout rate
						GRU	LSTM	GRU	LSTM	
8 h	CNN-LSTM	Adam	16	1	128	–	1	–	100	0.1
	CNN-GRU	Adam	64	1	128	2	–	100	–	0.2
24 h	CNN-LSTM	RMSprop	16	1	128	–	1	–	100	0.2
	CNN-GRU	Adam	32	1	128	2	–	100	–	0.2
72 h	CNN-LSTM	Adam	16	2	128	–	1	–	100	0.2
	CNN-GRU	Adam	32	1	64	2	–	100	–	0.2

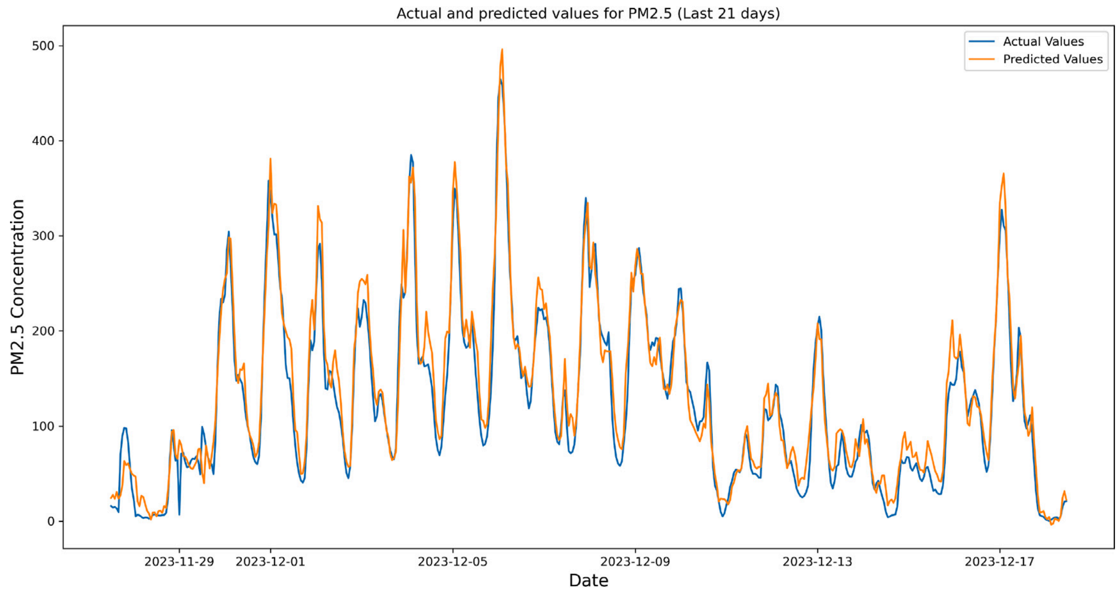


Fig. 13. Actual and predicted values for 21 days with CNN-LSTM for 8 h lag.

Table 3
Model performance comparison: General vs. Peak metrics.

Models	8 h			24 h			72 h		
	RMSE	R^2	MAE	RMSE	R^2	MAE	RMSE	R^2	MAE
General performance metrics									
LSTM	15.759	0.939	10.035	14.356	0.949	9.654	15.382	0.943	10.436
BiLSTM	15.727	0.939	10.217	15.731	0.939	11.111	15.481	0.942	10.349
GRU	15.087	0.944	9.932	15.249	0.943	10.454	15.872	0.939	11.079
BiGRU	15.471	0.941	10.487	15.123	0.944	10.613	16.057	0.937	10.907
CNN-LSTM	15.351	0.942	10.492	16.206	0.935	10.758	15.544	0.941	11.323
CNN-GRU	16.007	0.937	10.555	16.054	0.937	11.280	16.227	0.936	11.247
Peak performance metrics									
LSTM	36.014	0.660	28.960	29.305	0.774	23.330	31.180	0.743	24.566
BiLSTM	31.290	0.743	24.581	30.463	0.756	24.378	33.950	0.695	27.401
GRU	32.109	0.729	24.713	32.554	0.722	26.356	34.135	0.692	28.409
BiGRU	29.352	0.774	23.880	30.703	0.752	24.837	31.062	0.745	25.586
CNN-LSTM	28.156	0.792	22.454	30.832	0.750	24.333	28.605	0.784	22.704
CNN-GRU	29.625	0.770	23.598	28.948	0.780	23.957	33.571	0.702	27.455

both overall trends and extreme values well. Among them, CNN-LSTM and CNN-GRU models achieved the best performance in the 8 and 24 h time windows by providing the lowest errors in both general and peak value predictions. The variation in performance between different models across time windows can be attributed to several key factors. The GRU architecture's simplicity compared to other models provides an advantage, particularly in short-term forecasts where patterns are simpler. This is supported by the fact that more complex architectures such as LSTM and BiLSTM demonstrate better results as the forecast window extends to 24 and 72 h. Similarly, hybrid models such as CNN-LSTM and CNN-GRU are disadvantaged in short-term forecasts because they may not learn sufficiently when the patterns are predominantly temporal. The superiority of hybrid models in hybrid peak value prediction can be attributed to the spatial feature extraction capability provided by CNN layers. In particular, the CNN-LSTM model's best performance for peak value prediction across all time windows can be attributed to the combination of the CNN layers' local feature extraction capability and the LSTM layers' ability to model temporal dependencies. Sudden increases in air pollution concentration (peak values) are complex events that require both instantaneous features and temporal context, and hybrid models can simultaneously address these two requirements.

In conclusion, the CNN-LSTM model showed the best overall performance in terms of accuracy, efficiency, and predicting peak values. GRU and LSTM models also performed well, especially for short- and medium-term forecasts. These results show that it is important to consider not only average prediction accuracy but also how well models can predict extreme values. They also suggest that hybrid models have strong potential in time series forecasting.

For a better understanding of the prediction performance of the models and their ability to capture changes over time, a graph is given in Fig. 13 to show the PM2.5 predictions of the CNN-LSTM model using an 8-h time window on hourly data for the last 21 days. This model was selected because it showed strong performance both in general prediction accuracy and in forecasting peak values. When the agreement between the actual and predicted values is examined, it can be seen that the model successfully captures both peak concentrations and temporal fluctuations. The consistency, especially during sudden increases and decreases, indicates that the model effectively predicts short-term variations.

Similar to this study, there are various studies that estimate PM10 concentrations with daily data in Igdir city Tırınk and Öztürk (2023) obtained R^2 = 0.833 and RMSE = 35.575 values by using MARS and XGBoost algorithms for daily PM10 estimation. Şahin et al. (2020)

obtained R^2 = 0.881 and RMSE = 27.89 values with four-layer feed-forward artificial neural network (FFNN), and Batur and Babii (2024) reached RMSE = 12.54 by working on three-year daily data with Levenberg–Marquardt, Bayesian Regularization and Scaled Conjugate Gradient algorithms. These studies generally focused on daily PM10 estimations and reveal important contributions in this field. However, in this study, a more comprehensive data set was created by using hourly data for PM2.5 estimation and it was aimed to make estimations in more detailed time periods. Thus, the higher explanatory power of the models, the scope of the data set used, and the hourly estimation stand out as important elements that distinguish this study from similar studies in the literature.

5. Conclusion

This study discussed the application of various deep learning models to predict air pollution in Igdir province. The factors affecting air quality were examined in detail and the performance levels of different models in predicting PM2.5 levels were compared. The results showed that especially GRU and LSTM based models provided high accuracy in time series predictions. The predictions of the models generally followed the real values quite well and were successful in capturing daily trends. The findings obtained show that using deep learning models and hybrid versions of these models is an effective approach in air pollution prediction. In addition, it was observed that the use of different time windows affected the model performances and affected the prediction results.

Among the models tested, the GRU model achieved the best results in the 8-h time window with an R^2 of 0.944, RMSE of 15.09, and MAE of 9.93. While performance significantly improved in the 24-h window across all models, with the LSTM model showing the best results (R^2 = 0.949, RMSE = 14.36, MAE = 9.65), a slight increase in error was noted in the 72-h forecasts, where the BiLSTM model outperformed others (R^2 = 0.942, RMSE = 15.48). Additionally, CNN-LSTM and BiGRU hybrid models demonstrated competitive performance, especially in peak value predictions and longer-term forecasts. These results highlight the critical impact of time window selection and hyperparameter optimization on forecast accuracy.

This work offers a potential contribution to finding effective strategies to better cope with the chronic air pollution problem in Igdir province. In the future, it is recommended to increase the number of monitoring stations and integrate ecological and environmental factors more comprehensively in order to increase the prediction results. In addition, the use of different data sources and the development of hybrid model approaches will be important steps towards improving the prediction performance.

CRediT authorship contribution statement

Muhammed Kaya: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **İhsan Ömür Bucak:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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